

# Arduino IDE (offline): student guide

Write, check and upload Arduino programs on your Chromebook. Nothing to install, and no internet needed.

## 1 Open the IDE

- In Google Classroom, open **arduino-ide.html** and **download** it. It won't run inside the Google Drive preview.
- Open the **Files** app → **Downloads** and click **arduino-ide.html**. It opens in Chrome.
- Tip: move it to **My files** so it's easy to find next time.

## 3 Check and upload

- Plug the Arduino in with the USB cable.
- Click ☒ (Verify) to check your code. Mistakes show in red: click one to jump to that line.
- Click  (Upload). The first time, pick **Arduino Uno** in the list Chrome shows and click **Connect**.
- Wait for **Done uploading**.

## 2 Write your sketch

- Type your code in the editor. Your work is saved automatically in this browser.
- **Examples** has ready-made sketches, including one for every module in the sensor kit. Look for the HW-number printed on your board (for example `HW486_Photorresistor`).
- **Libraries** → **Include** adds a library's `#include` line for you.

## 4 See what your board says

- **Serial Monitor** tab → **Connect**. Choose the same speed as `Serial.begin()`, usually **9600 baud**.
- **Serial Plotter** tab draws numbers as a live graph. Name each line like this:  
`Serial.print("light:"); Serial.println(value);`

## 5 Save your work to Google Drive

- Autosave only keeps your work on **this** Chromebook. To keep it for real, click **Save As** and choose your **Google Drive** folder.
- After that, **Save** updates the same file. A sketch with several tabs is saved as a folder.
- To continue later, click **Open** and pick your file from Google Drive.

## ? If something goes wrong

- **The Arduino isn't in Chrome's list:** try another USB cable (some only charge) or another USB port.
- **"Couldn't open the port":** click **Disconnect** in the Serial Monitor and close any other tab using the board.
- **Upload failed:** unplug the board, plug it back in, and try again.
- **Nothing appears in the Serial Monitor:** make sure the baud rate matches `Serial.begin()`.
- **Red error you don't understand:** look at the line it points to, and the line just above it (often a missing `;`).

Shortcuts: **Ctrl+R** verify · **Ctrl+U** upload · **Ctrl+S** save · **Ctrl+Shift+M** Serial Monitor · **Ctrl+Shift+L** Serial Plotter